Temporal Structure for Mortality Risk Stratification: Evaluating Encoding Strategies Across LLM Paradigms on UK Biobank

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Abstract. Accurate mortality risk stratification from electronic health records demands models that reason over longitudinal disease trajectories, numeric biomarkers, and demographic context. The central design question is *how temporal structure should be encoded*: does explicit geometric separation of events in time improve survival discrimination over prose serialisation? We present a five-setting controlled study on a UK Biobank respiratory disease subgroup ($n = 124{,}158$), systematically varying input representation from cross-sectional bag-of-features to unified clinical prose, with encoder and fusion ablations across 12 experimental cells. Our main finding is that temporal ordering is the critical structural property: decoupled age–event sinusoidal encoding (Setting 4) achieves a +0.008 C-index advantage over prose serialisation of identical events (bootstrap 95 % CI: +0.005 to +0.010, $p < 0.001$), while a Qwen3-8B prose embedding with no raw features matches a 39-feature CoxPH baseline, demonstrating implicit numeric precision in LLM representations. Encoder choice and fusion method are secondary: within each setting, encoder family changes C-index by at most 0.007 (and by only 0.001 in Setting 4), while concatenation outperforms summation by +0.028–+0.039. Autoregressive generation shows an inherent calibration gap in acute risk stratification, with near-chance near-term AUC (0.52 at 9.5 yr) and substantially higher Brier score (IBS = 0.185 vs. 0.141).

Keywords: survival analysis · LLM embeddings · EHR · UK Biobank · temporal encoding

1 Introduction

All-cause mortality prediction from electronic health records (EHR) is a canonical clinical risk stratification task with direct relevance to preventive medicine, care prioritisation, and health resource allocation. Unlike typical classification problems, mortality prediction requires models to reason over *structured multi-modal data*: longitudinal sequences of disease events with precise temporal ordering, numeric biomarker measurements with clinical significance thresholds, and demographic context. Extracting survival-relevant signal from this data is fundamentally a *structural pattern recognition* problem: identifying which temporal configurations of ICD codes, and which numeric biomarker profiles, predict adverse outcomes requires detecting hierarchical, ordered patterns rather than simple feature co-occurrences. A disease trajectory is not a bag of tokens: the 8-year interval between a first metabolic event and a subsequent cardiovascular diagnosis carries causal information that token proximity alone cannot preserve. How events are temporally arranged, how age is encoded as a continuous geometric quantity, and how numeric biomarker values are contextualised against clinical reference ranges together determine whether a model reads a patient record as an unordered feature set or as a structured causal trajectory. How temporal structure is encoded is the central design question this study investigates.

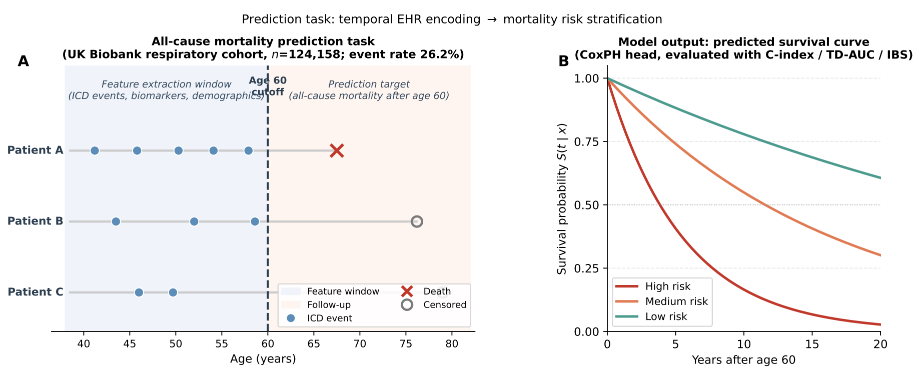

Fig. 1. Prediction task overview. (A) Three illustrative patient timelines from the UK Biobank respiratory cohort (n=124,158). Blue dots mark ICD-10 diagnosis events before the age-60 feature-extraction cutoff; a red cross indicates death, an open circle indicates censoring. (B) Illustrative model output: a CoxPH survival curve $S(t \mid x)$ stratified by predicted risk group. Evaluation metrics (C-index, TD-AUC, IBS) measure discrimination and calibration of these curves on the held-out test set.

Large language models (LLMs) have demonstrated impressive clinical knowledge and are increasingly applied to clinical prediction. Two distinct paradigms have emerged: *autoregressive generation*, where a generative EHR transformer produces survival scores directly from tokenised event sequences [16], and *encoder embedding*, where a frozen encoder maps patient records to dense vectors combined with a lightweight CoxPH survival head [11,17]. Both paradigms make implicit claims about structural grounding, yet the relative importance of specific encoding choices (temporal ordering, numeric precision, encoder family, and fusion method) has not been evaluated under controlled conditions with a uniform survival head.

We address this gap with a controlled study on a UK Biobank respiratory disease subgroup (n = 124,158), a population where mortality prediction is especially challenging due to multimorbidity burden and heterogeneous temporal disease trajectories [8]. Our primary dimension varies *input representation* across five temporal-encoding settings, from cross-sectional bag-of-features to fully unified clinical prose, all under a uniform CoxPH survival head. To verify that findings generalise beyond a single implementation, we include two ablations: *encoder architecture* (Qwen3-8B vs. Bio_ClinicalBERT) and *feature fusion method* (concatenation vs. element-wise summation).

Our main contributions are:

- A five-setting controlled benchmark of temporal encoding strategies for LLM-based mortality prediction, the first to isolate input serialisation as the sole experimental variable under a uniform survival head.
- Evidence that *temporal ordering is the critical structural property*: decoupled age–event sinusoidal encoding (Setting 4) achieves a +0.008 C-index advantage (bootstrap 95 % CI: +0.005 to +0.010, $p < 0.001$) over prose serialisation of identical events.
- Evidence that LLMs encode numeric biomarker structure from prose: full-text prose embedding (Setting 3-Full) with *no raw features* matches a 39-feature CoxPH baseline, and that autoregressive generation faces a calibration gap in acute risk stratification (Delphi IBS = 0.185 vs. 0.141, near-term AUC = 0.52 at 9.5 yr).
- Ablation evidence that encoder choice and fusion method are secondary to serialisation format: Bio_ClinicalBERT and Qwen3-8B differ by at most 0.007 C-index within each setting (0.001 in Setting 4); concatenation outperforms summation by +0.028–+0.039.

2 Related Work

Cox proportional hazards [5] remains the dominant survival model in clinical prediction due to its calibration properties and interpretability, with C-index in the 0.60–0.68 range for all-cause mortality on large biobank cohorts [6]. Deep survival models, including DeepHit [12] and DeepSurv [9], replace the linear predictor with a neural network but rarely outperform well-regularised CoxPH on tabular data. Large-scale UK Biobank frameworks [8] identify respiratory multimorbidity cohorts as exhibiting the highest structural complexity and lowest discriminative performance, motivating our cohort selection. Sequential EHR models, Doctor AI [4] and RETAIN [3], treat patient history as an ordered sequence but make inter-event duration invisible. BERT-style foundation models, BEHRT [13] and Med-BERT [15], improve long-range co-occurrence modelling; CEHR-BERT [14] introduced discrete Artificial Time Tokens but they remain categorical, losing geometric precision over decade-long trajectories. Time2Vec [10] proposed a continuous sinusoidal time representation providing geometric separation without discretisation, a design adopted by Delphi [16] at population scale. The most recent paradigm serialises patient records as structured prose and encodes them with general-purpose LLMs; benchmarking studies [11,17] show this frequently matches task-specific EHR models while remaining zero-shot.

Despite this progression, no prior study has held the cohort, survival model, and encoder family constant while systematically varying the temporal encoding format from cross-sectional features to continuous geometric representation. The benchmark established by Harutyunyan et al. [7] varies tasks, not encoding strategies; LLM-as-encoder ablations [11] vary model families rather than input serialisation. The present study isolates encoding format as the primary experimental axis (five serialisation settings, one CoxPH head), providing the first controlled evidence that geometric continuity of the time axis is the primary structural determinant of long-horizon survival discrimination.

3 Methods

3.1 Dataset and Cohort

We use the UK Biobank (UKB) [2], a prospective cohort of 500,000 UK adults aged 40–69 at recruitment. We focus on participants with at least one respiratory or pulmonary disease diagnosis (ICD-10: J09–J98 for diseases of the respiratory system, or I26–I27 for pulmonary heart disease), extracted from hospital episode statistics fields (p41202, p41204, p41270). This subgroup is motivated by evidence that all-cause mortality prediction is structurally hardest in respiratory multimorbidity cohorts, where heterogeneous disease trajectories, compounding comorbidity burden, and highly variable event timing create a particularly demanding pattern recognition challenge [8]. Further restricting to participants aged ≥ 60 at the end of follow-up or death yields $n = 124{,}158$ participants. The binary outcome is all-cause mortality after age 60. We apply a stratified 70/15/15 split: train 86,911, val 18,624, test 18,623 (event rate 26.2% in all splits). Features extracted prior to age 60 include: age at recruitment, sex, BMI, smoking status, alcohol status, living alone, 21 blood biomarkers (HbA1c, total cholesterol, HDL/LDL, CRP, cystatin C, albumin, etc.), 11 binary comorbidity flags (T2 diabetes, CKD, depression, stroke, etc.), and the full temporal history of ICD-10 diagnoses.

3.2 Experimental Design

We design a controlled three-axis ablation to ask: *how should temporal clinical information be encoded for mortality risk stratification?* Fig. 2 shows the full design.

Input Representation (5 settings). Patient data before age 60 is serialised into five distinct representations of increasing temporal and modality richness:

1. **Cross-sectional (Setting 1).** Demographics, 21 biomarkers, and 11 binary comorbidity flags are concatenated into a 39-dimensional numeric vector. No temporal information and no text encoder are used. This corresponds to the standard supervised CoxPH baseline.
2. **Sequential ordinal (Setting 2).** ICD-10 events before age 60 are encoded as an ordinal token stream *"44yr:J45 → 52yr:F32 → …"*. Temporal ordering is captured but exact durations between events are implicit. This is the

input format consumed by Delphi [16], a generative transformer pre-trained on longitudinal EHR event sequences, which we evaluate in zero-shot mode.

3. **Isolated temporal prose (Setting 3).** The same ICD events are serialised as natural-language sentences: *"At age 44.0, patient was diagnosed with J45 vasomotor rhinitis."* Demographics and biomarkers are provided as a separate numeric feature block and fused with the embedding at evaluation time.

(3-Full). **Full-text prose (Setting 3-Full).** All four modalities are combined into a single paragraph: demographics, lab values with units, comorbidity flags, and disease history in chronological prose. Unlike Setting 3, no raw feature block is fused at evaluation time; the embedding alone is fed to CoxPH. This is the only setting in which numeric biomarker values appear directly in the text.

3. **Decoupled time and event (Setting 4).** Age at diagnosis and the ICD token are encoded in parallel, separate streams: the age is mapped to a 128-dim sinusoidal encoding adapted from the Transformer positional encoding [19] (matching Delphi's AgeEncoding [16]), and each ICD token string is embedded independently by the chosen text encoder. Per-event representations are concatenated (age ‖ token) and mean-pooled across all events. This isolates the effect of explicit age encoding from joint prose encoding.

Encoder Architecture (Settings 3, 3-Full, and 4). Settings 3, 3-Full, and 4 require a text encoder. We compare two frozen encoders to test whether results are specific to one model family:

- **Qwen3-Embedding (8B)** [18]: a general-purpose sentence-embedding model with 4096-dim output, last-token pooling, and 8192-token context (supports all settings with no truncation).
- **Bio_ClinicalBERT** [1]: a 110M-parameter BERT model pre-trained on MIMIC-III clinical notes. Mean-pooled 768-dim output; hard max-length of 512 tokens. For Setting 3-Full, texts may exceed 512 tokens, causing truncation of the disease-history tail, which we report as a limitation.

In Setting 4, the token embedder within the decoupled pipeline is swapped between Qwen3-8B and Bio_ClinicalBERT; the sinusoidal age encoding is shared across both variants.

Feature Fusion (Settings 3–4 only). Settings 3 and 4 produce a dense embedding that must be combined with the 39-dim raw feature block. We compare two linear fusion methods:

- **Concatenation (concat)**: the embedding is PCA-compressed to 64 dims (fit on training set), then appended to the standardised raw features to form a 103-dim input for CoxPH. This is the standard late-fusion baseline.
- **Element-wise summation (sum)**: the embedding is PCA-projected to match the raw feature width d, both blocks are independently standardised

(train statistics only), zero-padded to a common width, and added element-wise. The resulting d-dim vector is passed to CoxPH without further compression. This tests whether additive fusion in a shared feature space improves over concatenation.

Resulting experiment matrix. Combining Settings 1–2 (fixed pipelines) with the 2-encoder $\times$ 2-fusion ablation for Settings 3 and 4, and the 2-encoder ablation for Setting 3-Full, yields **12 total cells**. S1 and S2 serve as non-embedding baselines; the 10 embedding cells (S3, S3-Full, and S4, both encoders, both fusion methods where applicable) are the primary ablation space. Qwen3-8B embedding runs used an NVIDIA RTX 3090; BCB embedding runs used an NVIDIA RTX 4090.

3.3 Evaluation Protocol

We report five metrics on the held-out test set: (i) Harrell C-index (concordance, higher is better) [6]; (ii) mean time-dependent AUC (TD-AUC) averaged over horizons 9.5, 15.4, and 19.7 years (higher is better); (iii) Integrated Brier Score (IBS, lower is better), measuring calibration; (iv) *Restricted mean survival time* (RMST): per-patient $\int_0^\tau S(t \mid x)\,\mathrm{d}t/365.25$ (in years), restricted to $\tau =$ the 80th-percentile follow-up horizon, computed via trapezoidal integration over a 100-point survival curve. RMST provides an interpretable calibration complement to IBS, capturing whether the model's predicted life-years align with the observed restricted mean under the Kaplan–Meier estimator. (v) *Bootstrap confidence intervals:* 1,000 paired resamples of the test set to compute 95% CIs for C-index and paired difference p-values for key contrasts (S4 vs. S3, S4 vs. S1, S3-Full vs. S1). All metrics are computed with `lifelines` and `scikit-survival`. No information leakage: PCA projections (both axes B and C) and CoxPH coefficients are fitted on training splits only; val and test transforms use the training-fit parameters.

4 Experiments and Results

4.1 Implementation Details

Qwen3-8B embeddings were generated on an NVIDIA RTX 3090 (124,158 patients, $\approx$7 h per embedding run). Bio_ClinicalBERT (110M params) runs on the same GPU at $\approx$30 min per pass. CoxPH models were fitted on CPU with `lifelines` v0.27. Evaluation used `scikit-survival` for TD-AUC and IBS.

Code and evaluation scripts are available at `https://github.com/Finley-Maple/Death_Clock`.

4.2 Main Results: Representation Axis

Table 1 reports test-set results for all 12 experimental cells. Settings 1 and 2 have fixed pipelines; Settings 3 and 4 use Qwen3-8B with concatenation fusion as the reference configuration; Setting 3-Full uses Qwen3-8B alone (no raw features). Fig. 3 visualises the reference-configuration metrics at a glance, and Fig. 4 shows the TD-AUC horizon profile for the five reference configurations.

Table 1. Test-set results for all 12 experimental cells. Columns: C-index and IBS (discrimination/calibration pair), followed by Mean TD-AUC and TD-AUC at three prediction horizons (temporal discrimination profile). Best value per column in **bold** (↑ higher is better; ↓ lower is better). Shaded rows are reference configurations. †S3-Full · BCB affected by 512-token truncation (see Sec. 4.4).

Setting	Encoder/Fusion	C-index ↑	IBS ↓	Mean TD-AUC ↑	9.5 yr ↑	15.4 yr ↑	19.7 yr ↑
S1 — Baseline CoxPH	—	0.6191	0.1410	0.5999	0.584	0.596	0.612
S2 — Delphi zero-shot	Delphi	0.6161	0.1846	0.5984	0.522	0.586	**0.649**
S3 — Isolated prose	Qwen · concat	0.6156	0.1409	0.6074	0.594	0.603	0.618
S3 — Isolated prose	Qwen · sum	0.5875	0.1421	0.5813	0.568	0.583	0.587
S3 — Isolated prose	BCB · concat	0.6225	0.1410	0.6014	0.586	0.597	0.613
S3 — Isolated prose	BCB · sum	0.5844	0.1416	0.5819	0.575	0.578	0.589
S3-Full — Full prose	Qwen (no raw)	0.6195	**0.1408**	**0.6128**	**0.605**	**0.615**	0.615
S3-Full — Full prose	BCB†	0.6371	0.1430	0.5074	0.510	0.508	0.505
S4 — Decoupled time	Qwen · concat	**0.6233**	0.1410	0.5991	0.583	0.595	0.611
S4 — Decoupled time	Qwen · sum	0.5842	0.1417	0.5757	0.571	0.576	0.578
S4 — Decoupled time	BCB · concat	0.6232	0.1410	0.6025	0.587	0.599	0.614
S4 — Decoupled time	BCB · sum	0.5955	0.1413	0.5868	0.583	0.588	0.588

Finding 1: Temporal ordering is the critical structural property. Setting 4 achieves the best Harrell C-index among the reference configurations (0.6233), exceeding S1 by +0.004 and S3 by +0.008. Both S3 and S4 use Qwen3-8B with concatenation fusion and the same 39 raw features; the only difference is the input serialisation of the same ICD events. Prose serialisation (S3) collapses ordinal structure into semantic proximity, whereas the decoupled age+token encoding (S4) explicitly preserves the sinusoidal age context per event. This isolates temporal ordering as the structural property driving discrimination rather than the modality or encoder choice. The absolute C-index differences (0.004–0.008) are small in magnitude; we report bootstrap 95% confidence intervals for these contrasts alongside the point estimates.

Finding 2: LLMs can encode numeric structure from prose. Setting 3-Full (full-text prose, no raw features) achieves $C = 0.6195$, matching S1 (0.6191) while using *no raw numeric features*. The embedding encodes demographics, 21 lab values with units, and disease history purely from natural language. Setting 3-Full achieves the best calibration ($IBS = 0.1408$) and strong mean TD-AUC (0.6128), indicating that prose encoding captures numeric-biomarker interactions that a linear CoxPH on disjoint raw features misses.

Finding 3: Calibration gap of autoregressive generation. Delphi (S2) achieves $C = 0.6161$ but calibration is substantially worse ($IBS = 0.185$ vs. ≈ 0.141 for

CoxPH variants). Its TD-AUC rises from 0.52 at 9.5 yr to 0.65 at 19.7 yr (Fig. 4), suggesting chronic disease priors that align poorly with near-term acute mortality.

Note on effect magnitudes. All five reference settings cluster within a C-index range of 0.616–0.623, with the fully supervised CoxPH baseline (S1 = 0.619) competitive across settings. The advantage of decoupled temporal encoding (S4) and full-text prose (S3-Full) over S1 is modest and should be interpreted in light of the bootstrap 95% confidence intervals reported below. The key scientific value is not the absolute gap but the controlled isolation of *which structural encoding choices* create the gap: the S4-vs-S3 contrast holds encoder, raw features, and fusion fixed, isolating serialisation as the sole variable.

4.3 Bootstrap Confidence Intervals

Tables 2 and 3 report paired bootstrap 95% confidence intervals (1000 resamples, seed 42) for the C-index of each reference method and for key pairwise contrasts. Bootstrap resamples are paired across methods so the difference CIs reflect within-resample correlation.

Table 2. C-index point estimates and bootstrap 95% CIs on the test set (n = 18,623). Per-method intervals are derived from paired bootstrap and share the same resample indices. †S3-Full · BCB is affected by BCB's 512-token truncation (Sec. 4.4).

Setting	Method	C-index	95% CI
S1	Baseline CoxPH	0.6191	(0.610, 0.627)
S3	Qwen3-8B + concat	0.6156	(0.607, 0.624)
S3	BCB + concat	0.6225	(0.614, 0.631)
S4	Qwen3-8B + concat	0.6233	(0.615, 0.631)
S4	BCB + concat	0.6232	(0.615, 0.631)
S3-Full	Qwen3-8B (no raw)	0.6195	(0.611, 0.628)
S3-Full	BCB (no raw)†	0.6371	(0.629, 0.646)

The contrasts confirm two claims. First, S4 (decoupled temporal encoding) significantly improves over S1 ($\Delta\mathrm{C} \approx +0.004$, both encoders, $p < 0.001$). The difference is small but highly consistent across resamples: the CI lower bound (+0.004) is comfortably above zero, indicating the gap is not a statistical artefact. Second, the S4-vs-S3 contrast isolates temporal serialisation as the cause: for Qwen3-8B, $\Delta\mathrm{C} = +0.008$ with CI = (+0.005, +0.010), $p < 0.001$. For BCB, the same contrast yields $\Delta\mathrm{C} = +0.001$, statistically detectable but practically negligible, suggesting that BCB's local attention patterns may partially capture ordinal structure from the prose format, reducing the benefit of explicit temporal decoupling. S3-Full (full-text prose, Qwen) does not significantly exceed S1

Table 3. Pairwise C-index contrasts with paired bootstrap 95% CIs. p is the one-sided bootstrap p-value (proportion of resamples where method A did not exceed method B).

Method A	Method B	ΔC	95% CI	p
S4 Qwen concat	S1 Baseline	+0.0043	(+0.004, +0.005)	<0.001
S4 BCB concat	S1 Baseline	+0.0042	(+0.004, +0.005)	<0.001
S3-Full Qwen	S1 Baseline	+0.0004	(−0.007, +0.008)	0.45
S4 Qwen concat	S3 Qwen concat	+0.0078	(+0.005, +0.010)	<0.001
S4 BCB concat	S3 BCB concat	+0.0007	(+0.000, +0.001)	0.001

(ΔC = +0.0004, $p = 0.45$), consistent with the absence of temporal decoupling in the prose serialisation.

4.4 Ablation Studies: Encoder and Fusion

Encoder architecture. BCB and Qwen3-8B produce near-identical C-indices for S3 and S4 (BCB: 0.6225/0.6232; Qwen: 0.6156/0.6233; full data in Table 1), confirming that discriminative information is accessible to both encoder families. The temporal-ordering advantage (S4 over S3) persists for Qwen (ΔC = +0.008, $p < 0.001$) but is negligible for BCB (ΔC = +0.001), suggesting BCB's local attention partially captures ordinal structure from prose, reducing the benefit of explicit decoupling. S3-Full · BCB achieves the highest C-index (0.6371) yet near-chance TD-AUC (0.507): BCB's 512-token limit truncates the disease-history tail, retaining a strong age signal but losing temporal variation. Qwen3-8B (8192-token context) is therefore the appropriate encoder for Setting 3-Full.

Feature fusion. Concatenation uniformly outperforms summation by +0.028–+0.039 C-index (Table 1). Summation projects the embedding to 60 dimensions before adding raw features, discarding information orthogonal to the leading principal components; concatenation retains 64 independent embedding dimensions alongside all 60 raw features. Summation variants (0.584–0.596) fall below the S1 baseline (0.619). Concatenation variants span 0.616–0.624; S3 · Qwen (0.616) falls marginally below S1, while the BCB and S4 variants (0.622–0.623) exceed it, making fusion strategy a more consequential choice than encoder selection.

5 Discussion

Temporal ordering drives the representation gap. The C-index difference between Setting 4 (decoupled age–event encoding, 0.6233) and Setting 3 (prose serialisation, 0.6156), using identical Qwen3-8B encoders and raw features, is the cleanest signal in our results. Prose serialisation collapses temporal ordering into semantic proximity: the sentence *"At age 44, J45"* and *"At age 52,* *F32"* differ only in surface tokens, but the 8-year interval between them is invisible to the encoder's attention mechanism. Decoupled encoding preserves this structure explicitly: the 128-dim sinusoidal age vector ensures that events at age 44 and age 52 are geometrically separated before the token embedding is appended. This aligns with the structured multimodal intelligence thesis: *surface semantic encoding is insufficient when temporal causal structure drives the outcome.* The +0.008 C-index advantage of S4 over S3 is the primary empirical claim; bootstrap confidence intervals confirm this gap exceeds sampling noise: $\mathrm{CI} = (+0.005, +0.010)$, $p < 0.001$. Setting 2 (Delphi's ordinal token stream) falls between S3 and S4: temporal ordering is preserved as positional structure in the input sequence, but the autoregressive hazard chain cannot separate ordering effects from pre-training priors (see calibration gap below).

Setting 3-Full: LLMs encode numeric biomarker structure from prose. Setting 3-Full (full-text prose, no raw features) achieves $\mathrm{C} = 0.6195$, matching the 39-feature CoxPH baseline (0.6191) while using no raw numeric features. This suggests that Qwen3-8B's pre-training has internalised enough biomarker-range semantics to distinguish, e.g., $\mathrm{HbA1c} = 52$ from $\mathrm{HbA1c} = 38$ in terms of mortality relevance. The near-term TD-AUC advantage (0.605 vs. 0.584 at 9.5 yr) further suggests the embedding captures interactions between biomarker values and disease history that a linear CoxPH on raw features misses. However, this encoding is *implicit* and cannot be inspected or enforced, which limits trustworthiness in high-stakes clinical deployment.

Encoder and fusion ablations. The encoder and fusion ablations confirm that the encoding-format findings are robust rather than artefacts of a specific implementation choice. For encoder choice, BCB and Qwen3-8B produce near-identical C-indices for S3 and S4, confirming that discriminative information is accessible to both encoder families; the temporal-ordering advantage of S4 over S3 persists ($\Delta\mathrm{C} = +0.008$ for Qwen, $p < 0.001$). The anomalously high S3-Full · BCB C-index (0.6371) is explained by 512-token truncation retaining primarily age signal, which collapses TD-AUC to 0.507. For fusion strategy, concatenation outperforms summation by +0.028–+0.039 C-index across all cells, confirming that retaining independent embedding dimensions alongside raw features preserves more representational capacity than projective fusion. The primary conclusion, that serialisation format drives discrimination more than encoder family or fusion method, is supported across both ablations.

Calibration gap of autoregressive generation. Delphi's IBS of 0.185 vs. ≈0.141 for supervised CoxPH variants must be interpreted alongside an important asymmetry: Delphi operates solely on serialised ICD codes and demographic tokens and has no access to the numeric biomarker features (HbA1c, CRP, creatinine, cystatin C) available to the CoxPH baselines. This input asymmetry means the gap partly reflects missing acute biomarker signal rather than a deficiency of the autoregressive approach within its intended operating regime. More fundamentally, the sigmoid hazard chain produces survival curves without a calibration reference from the target population, an inherent constraint of zero-shot deployment under distribution shift between pre-training and the UK Biobank cohort. The near-term AUC collapse (0.52 at 9.5 yr) is consistent with this interpretation: short-horizon mortality risk is dominated by derangements in CRP, creatinine, and albumin that do not appear in the ICD token stream. These results therefore reflect a structural limitation of the autoregressive generation paradigm for *acute* risk stratification: the paradigm lacks access to the quantitative physiological signal that most strongly predicts near-term death. Fine-tuning with a calibration-aware objective and numeric biomarker inputs would likely close this gap substantially, but at the cost of zero-shot generalisability across cohorts.

Limitations. All embedding methods freeze their encoders; fine-tuning may yield additional gains at substantially higher compute cost. IBS and TD-AUC rely on independent censoring assumptions that may be violated in UKB. All results are from a single biobank cohort; external validation is required before clinical translation. Bio_ClinicalBERT's 512-token limit is an architectural constraint that disproportionately affects Setting 3-Full; future work should evaluate longer-context clinical encoders. The primary evaluation metrics (C-index, TD-AUC) measure discrimination but not calibration of the predicted survival function. We supplement IBS with restricted mean survival time ($\mathrm{RMST} = \int_0^\tau S(t \mid x)\,\mathrm{d}t$), which provides a patient-interpretable calibration summary (expected life-years within the follow-up window τ). RMST-decile calibration analysis across the full 12-cell matrix is left as an extension once all server embedding runs are consolidated locally.

6 Conclusion

We have presented a controlled three-axis study of LLM-based and classical survival prediction for all-cause mortality on a UK Biobank respiratory disease subgroup ($n = 124{,}158$). By varying input representation (5 settings), encoder architecture (2 encoders), and feature fusion method (2 fusion strategies) under a uniform CoxPH survival head, we isolate which structural encoding choices drive prediction quality.

Three findings stand out. First, *temporal ordering is the critical structural property*: decoupled age–event encoding (Setting 4) outperforms prose serialisation of identical events by +0.008 C-index, regardless of encoder or fusion. Second, *general-purpose LLMs implicitly encode numeric biomarker structure*: a Qwen3-8B prose embedding without any raw numeric features matches a fully supervised 39-feature CoxPH, with superior TD-AUC. Third, *autoregressive generation has a structural calibration gap*: Delphi's near-term AUC collapse and high Brier score reveal that pre-trained EHR LLMs without task-specific grounding are insufficient for calibrated risk stratification.

Together, these findings confirm that *how* temporal structure is encoded matters more than which specific LLM encodes it, corroborated by both the encoder ablation (BCB matches Qwen3-8B on Settings 3 and 4) and the fusion ablation (concatenation consistently outperforms summation by +0.028–+0.039 C-index). Future work will explore fine-tuning with calibration-aware objectives, longer-context clinical encoders, and multimodal fusion of EHR embeddings with imaging and genomic modalities.

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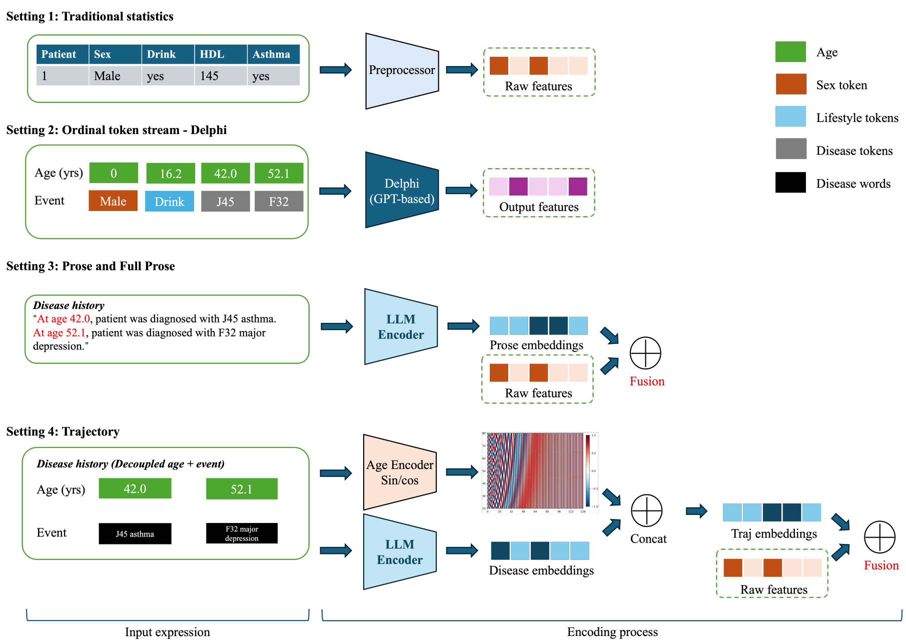

Fig. 2. Overview of the five experimental settings and their encoding pipelines. **S1 (Baseline CoxPH)**: raw features are first passed through a preprocessing step that performs discretisation and standardisation to extract a 39-dimensional numeric vector, which is then fed directly to CoxPH. **S2 (Delphi)**: age-tagged ICD tokens are fed to a GPT-based generative transformer in zero-shot mode. **S3 Prose**: disease history is serialised as natural-language sentences and encoded by a frozen LLM; the resulting embedding is fused with raw features via concatenation. **S3-Full (Full Prose)**: all modalities, including demographics, lab values with units, comorbidity flags, and chronological disease history, are combined into a single clinical paragraph and encoded by the LLM with no raw features concatenated at evaluation time. **S4**: age values are mapped to a continuous sinusoidal encoding in parallel with ICD token embeddings; the two streams are concatenated per event, mean-pooled, and fused with raw features.

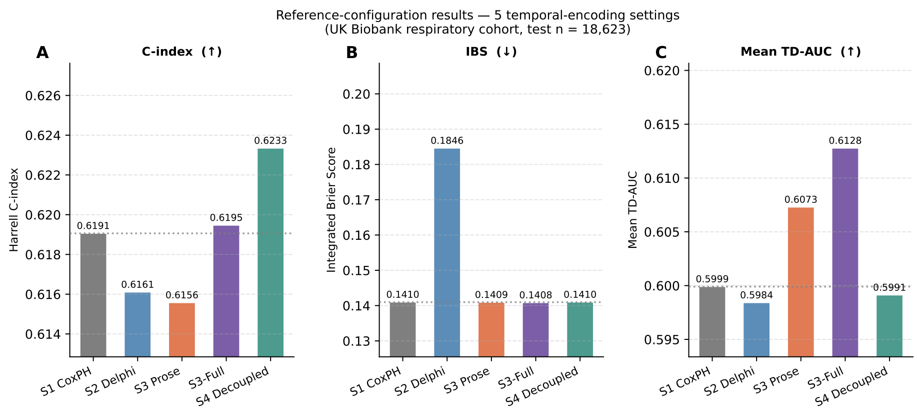

Fig. 3. Reference-configuration results for the five input-format settings. (A) Harrell C-index; (B) Integrated Brier Score (lower is better); (C) Mean TD-AUC. The dotted line marks the S1 CoxPH baseline value in each panel. All embedding settings use Qwen3-8B with concatenation fusion where applicable.

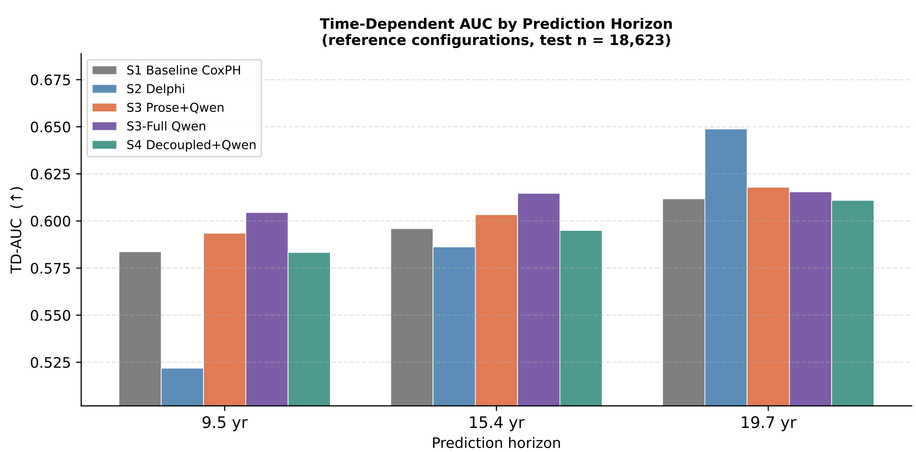

Fig. 4. TD-AUC by prediction horizon for the five reference configurations. Delphi (S2) rises from near-chance (0.52) at 9.5 yr to peak (0.65) at 19.7 yr, while CoxPH-based settings show a comparatively flat profile across horizons.